Bass Model Exercise Solutions

We have provided you with *Gleacher Game Bass Model* spreadsheet that calculates and plots daily customer arrivals over time for the markets in the game. The spreadsheet uses the information in the *Gleacher Game Case Brief* and the "p" and "q" parameters provided in the market research reports. Please review the spreadsheet and make sure that you understand its construction.

Answer all questions in this section for the MD Cancer - Bladder & Kidney market, assuming initial market size = 15,000, mean of WTP = $1,300, standard deviation of WTP = $130, materials cost per unit = $375, manufacturing overhead per unit = $80, and shipping cost per unit = $20. Assume that the forecast period starts on day 1. Brief numerical answers are sufficient.

For questions 3-4, assume a retail price of $900 per unit and $0 in advertising spending. Round your answers to two decimal places.

1. How many Innovator customers will arrive on the following days?

   Day 1: 3.00 Day 364: 2.60 Day 728: 1.72

2. How many Imitator customers will arrive on the following days?

   Day 1: 0 Day 364: 6.03 Day 728: 12.83

For questions 5-7, assume a retail price of $900 per unit and $3,000 per day in advertising spending in the first 364 days. Round your answers to two decimal places.

3. How many Innovator customers will arrive on the following days?

   Day 1: 3.00 Day 364: 1.25 Day 728: 0.46

4. How many Imitator customers will arrive on the following days?

   Day 1: 0 Day 364: 12.75 Day 728: 6.77

5. How many Advertising customers will arrive on the following days? Advertising customers are customers attracted through advertising.

   Day 1: 18.00 Day 364: 7.48 Day 728: 0

6. Assuming no advertising, if you were able to satisfy all of the arriving customers in this product market, i.e., 100% of arriving customers buy from you, how many units would you sell over the next four years? What retail price would you have to charge to make that happen?

   13,824 units

   Retail price = $910 or less

7. Assuming a retail price of $1,200 and potential competitor activity, do you think that you will be able to satisfy all potential customers in this product market? Why or why not? What would you have to do to satisfy more customers? Would you want to do that? When would you do that? A qualitative answer is fine here.

   Some customers will not pay $1,200, so you would not be able to satisfy all potential customers. Competitors might also undercut your price and take market share.

   To satisfy all customers, you would need to sell at $910, which is the minimum willingness to pay in the market based on the assumptions provided. This would help build up brand loyalty in order to attract more imitators in the future, but you would be making less profit on every sale in the meantime. You might do it to make it less attractive for competitors to enter the market, or to compete on price if forced to do so by existing competition.

8. What would be your cumulative contribution at the end of Year 4 in each of the following scenarios? Round your answers to the nearest thousand dollars.

   a) Retail price of $900 and $0 in daily advertising: $3,249K

   b) Retail price of $900 and $3,000 in daily advertising for the first 364 days: $2,391K

   c) Retail price of $1,200 and $0 in daily advertising: $4,597K

   d) Retail price of $1,200 and $3,000 in daily advertising for the first 364 days: $4,287K

9. Under which circumstances would you advertise a product? Under which circumstances would you not advertise a product? A qualitative answer is fine here.

   Team specific.

   Depending on the number of customers directly attracted by advertising and the product's contribution margin per unit, the incremental contribution from those customers may or may not cover the incremental cost of attracting them. However, advertising builds up a product's brand loyalty to speed up the arrival of imitators in the future, so the investment can pay off over time.

   Strategically, advertising can be used to stave off the entry of a competitor or to avoid a price war.

It is best to advertise early in the lifecycle for both the market and your product, when most arriving customers are innovators and the effect on future imitators is highest. It makes much less sense to advertise later in the market's or product's lifecycle, when few customers are left in the market and/or most of the demand for your product comes from imitators.

Customers come on the day that the advertising dollars are spent, so do not advertise when you don't have capacity to satisfy the resulting demand.

Do not advertise an unprofitable product.